DAY 5 – SIEM (Security Information and Event Management)

Objective

The objective of this session is to develop a comprehensive understanding of **SIEM (Security Information and Event Management)**, its architecture, workflow, and its critical role in modern **Security Operations Centers (SOC)**.

By the end of this session, learners will be able to:

- Understand SIEM concepts and core components
- Explain how SIEM processes and analyzes logs
- Identify real-world use cases in SOC environments
- Gain awareness of popular SIEM tools used in the industry

1. What is SIEM?

SIEM (Security Information and Event Management) is a centralized security solution that collects, aggregates, correlates, and analyzes log data from multiple sources in real time to detect and respond to security threats.

It combines two key functions:

- **SIM (Security Information Management):** Log collection and storage
- **SEM (Security Event Management):** Real-time monitoring and alerting

Importance in Cybersecurity

SIEM plays a critical role in:

- Continuous security monitoring
- Threat detection and analysis
- Incident investigation and response
- Compliance and auditing

2. Core Components of SIEM

A SIEM system consists of several essential components:

1. Log Collection

- Collects logs from various sources such as:
  - Servers
  - Firewalls
  - Endpoints
  - Network devices

2. Log Storage

- Stores logs securely for:
  - Historical analysis
  - Compliance requirements
  - Forensic investigations

3. Correlation Engine

- Analyzes logs and identifies patterns
- Detects suspicious behavior across multiple systems

4. Alerting System

- Generates alerts when predefined rules are triggered
- Notifies SOC analysts about potential threats

5. Dashboard / Visualization

- Provides a graphical interface
- Displays alerts, logs, and system status in real time

3. How SIEM Works:

The SIEM workflow follows a structured process:

Step 1: Log Collection

Logs are collected from multiple systems and devices across the network.

Step 2: Normalization & Storage

Logs are converted into a standard format and stored for analysis.

Step 3: Correlation

The SIEM applies rules to identify patterns and relationships between events.

Step 4: Detection

Suspicious or abnormal activities are detected based on correlation rules.

Step 5: Alert Generation

Alerts are generated and forwarded to SOC analysts for investigation.

4. Real-World Example

Scenario: Brute Force Attack Detection

- Multiple failed login attempts occur across different servers
- SIEM collects logs from all systems
- Correlation engine identifies repeated failures from the same IP
- SIEM triggers a **brute force alert**
- SOC analyst investigates and takes action

5. Popular SIEM Tools

1. Splunk

Overview:
Splunk is one of the most widely used SIEM platforms, known for its powerful search, analytics, and real-time monitoring capabilities.

Key Features:

- Real-time log collection and indexing
- Advanced search using SPL (Search Processing Language)
- Interactive dashboards and visualizations
- Alerting and incident detection
- Machine learning-based threat detection

Advantages:

- Highly scalable for large enterprise environments
- Fast search and analysis of large datasets
- Strong community support and integrations

Consideration:

- Licensing cost can be high compared to open-source tools

Use Case:

Detecting suspicious login patterns and creating alerts based on abnormal behavior across multiple systems.

2. ELK Stack (Elasticsearch, Logstash, Kibana)

Overview:
The **ELK Stack** is a popular open-source SIEM solution consisting of three main components.

Components:

- **Elasticsearch** – Stores and indexes log data for fast search
- **Logstash** – Collects, processes, and forwards logs

- **Kibana** – Provides dashboards and visualizations

Key Features:

- Open-source and customizable
- Real-time log ingestion and analysis
- Powerful visualization through dashboards
- Flexible data processing

Advantages:

- Cost-effective (free and open-source)
- Highly customizable based on requirements
- Strong community and ecosystem

Consideration:

- Requires configuration and maintenance effort
- Less out-of-the-box compared to commercial SIEMs

Use Case:

Monitoring logs across multiple servers and visualizing attack patterns using dashboards.

3. IBM QRadar

Overview:
IBM's IBM QRadar is an enterprise-grade SIEM platform known for its strong correlation and threat detection capabilities.

Key Features:

- Real-time event and flow analysis
- Advanced correlation engine
- Built-in threat intelligence integration
- Offense-based alerting system
- Automated incident prioritization

Advantages:

- Strong threat detection and correlation
- Efficient alert prioritization reduces noise
- Suitable for large enterprise SOC environments

Consideration:

- Requires training to use effectively
- Higher cost for deployment and maintenance

Use Case:

Detecting complex attacks involving multiple stages, such as lateral movement and data exfiltration.

Comparison

Feature	Splunk	ELK Stack	IBM QRadar
Type	Commercial	Open-source	Commercial
Ease of Use	High	Medium	Medium
Customization	High	Very High	Moderate
Cost	Expensive	Free	Expensive
Best For	Large enterprises	Flexible setups	Enterprise SOC

6. SOC Use Cases of SIEM

1. Detecting Brute Force Attacks

A brute force attack involves multiple login attempts using different password combinations.

How SIEM Helps:

- Collects authentication logs from multiple systems
- Detects repeated failed login attempts
- Correlates attempts from the same IP or user
- Generates alerts when a threshold is exceeded

Example:

50 failed login attempts from a single IP in 1 minute → Alert triggered

SOC Action:

- Block IP address
- Reset affected account
- Investigate attack source

2. Identifying Malware Activity

Malware can generate unusual system behavior such as unknown processes or suspicious connections.

How SIEM Helps:

- Monitors endpoint logs and system processes
- Detects abnormal activity (unknown executable, unusual traffic)
- Correlates events across systems

Example:

A system suddenly starts communicating with a suspicious external IP

SOC Action:

- Isolate infected system
- Run malware analysis
- Remove malicious files

3. Monitoring Insider Threats

Insider threats come from authorized users misusing access.

How SIEM Helps:

- Tracks user activity and access logs

- Detects unusual behavior (large data transfers, odd login times)
- Correlates actions across systems

Example:

Employee downloads large sensitive files at midnight

SOC Action:

- Investigate user activity
- Restrict access
- Report incident

4. Investigating Security Incidents

SIEM helps analysts understand what happened during a security event.

How SIEM Helps:

- Provides centralized log visibility
- Correlates events from multiple sources
- Reconstructs attack timeline

Example:

Suspicious login → privilege escalation → data access

SOC Action:

- Analyze logs
- Identify attack path
- Implement corrective measures

5. Real-Time Threat Detection & Response

SIEM enables continuous monitoring and immediate detection.

How SIEM Helps:

- Monitors logs in real-time
- Applies correlation rules instantly
- Triggers alerts without delay

Example:

Unusual traffic spike detected → possible DDoS attack

SOC Action:

- Activate response plan
- Block malicious traffic
- Monitor system stability

Conclusion

SIEM plays a critical role in modern SOC environments by:

- Centralizing log data
- Detecting threats in real time
- Enabling faster incident response
- Improving overall security visibility

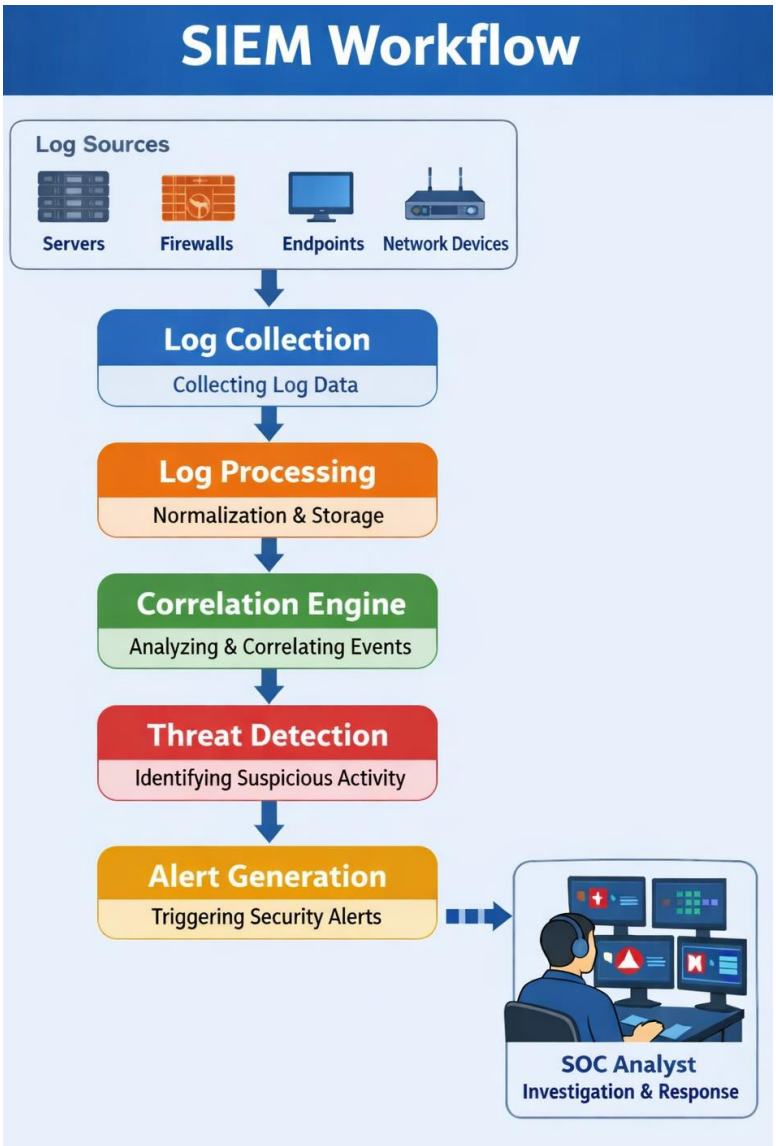
Understanding SIEM workflows and use cases is essential for any aspiring **SOC Analyst**, as it forms the backbone of security monitoring and threat detection.

7. Class Activity

Task:

Create a **SIEM Workflow Diagram** showing:

- Log collection
- Log processing
- Correlation
- Alert generation



Objective:

To visually understand how SIEM processes data from input to detection.

8. Assignment

1. Explain SIEM Workflow

SIEM workflow involves collecting logs, normalizing data, applying correlation rules, detecting suspicious activities, and generating alerts for further investigation by SOC analysts.

2. List 3 SIEM Tools

- Splunk
- ELK Stack
- IBM QRadar

3. Real-World Detection Example

A SIEM system detects multiple failed login attempts from the same IP address across different systems within a short time. This pattern indicates a possible brute force attack, triggering an alert for investigation.

Conclusion

This session provided a strong foundation in understanding how SIEM systems operate within a SOC environment. SIEM plays a vital role in **centralized monitoring, threat detection, and incident response**, making it an essential tool for cybersecurity professionals.

A solid understanding of SIEM will support advanced topics such as:

- Threat hunting
- Incident response
- Security monitoring
- Digital forensics